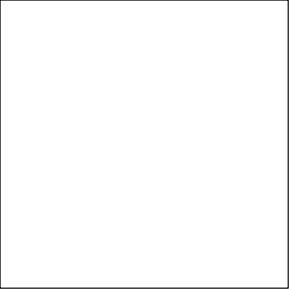
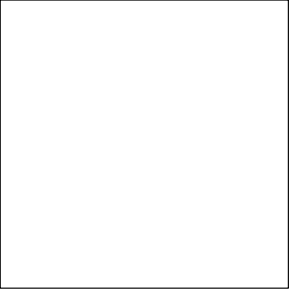


Microcontroller



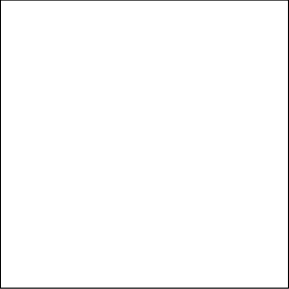
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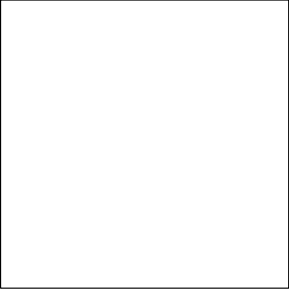
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RF

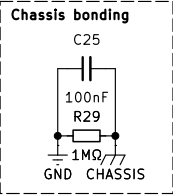


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Power



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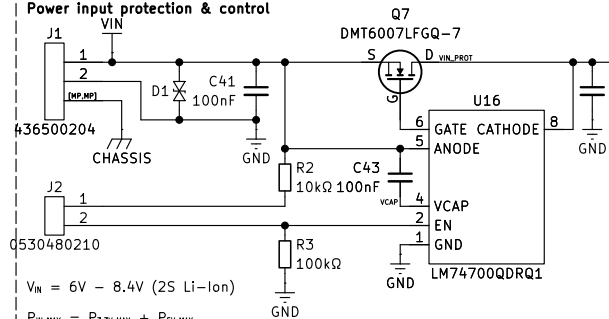


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Power input protection & control

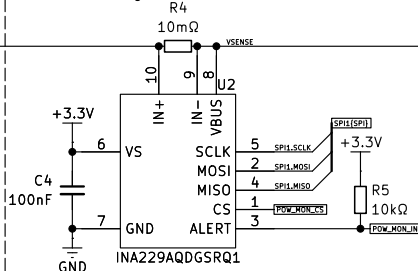


$$V_{IN} = 6V - 8.4V \text{ (2S Li-Ion)}$$

$$P_{IN\text{ MAX}} = P_{3.3V\text{ MAX}} + P_{5V\text{ MAX}} + P_{SENSE\text{ MAX}} + P_{PYRO\text{ CH MAX}} = 11.65W + 16.67W + 0.34W + 6.22W = 34.88W$$

$$I_{IN\text{ MAX}} = P_{IN\text{ MAX}} / V_{IN\text{ MIN}} = 34.77W / 6V = 5.81A$$

Power monitoring



$$V_{SENSE\text{ MAX}} = 163.84mV$$

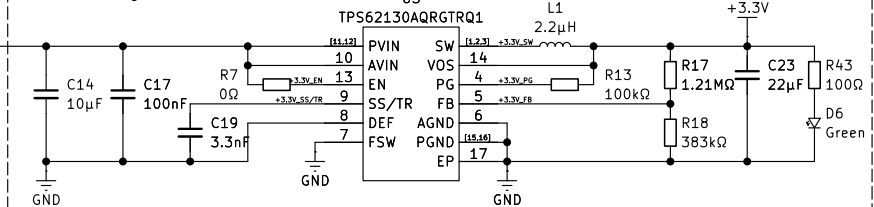
$$R_{SHUNT\text{ MAX}} = V_{SENSE\text{ MAX}} / I_{MAX} = 163.84mV / (34.54W / 6V) = 28.46mΩ = 10mΩ$$

$$P_{MAX} = R_{SHUNT} \times I_{MAX}^2 = 10mΩ \times 5.79A^2 = 0.34W$$

Pyrotechnic channels switch

J3
0530480210

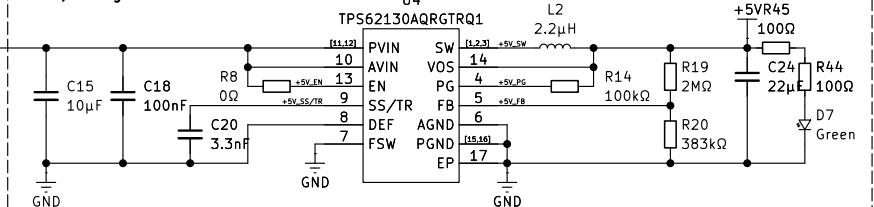
3.3V DC/DC Regulator



$$V_{OUT} = ((R_{TOP} / R_{BOTTOM}) + 1) \times 0.8V_D = (V_{IN} - V_F) / R = ((1.21MΩ / 383kΩ) + 1) \times 0.8V_{3.3V} - 2.2V / 100Ω = 11mA$$

$$P_{IN\text{ MAX}} = (P_{3.3V} / \eta_{3.3V @ 3A}) = ((3.3V \times 3A) / 0.85) = 11.65W$$

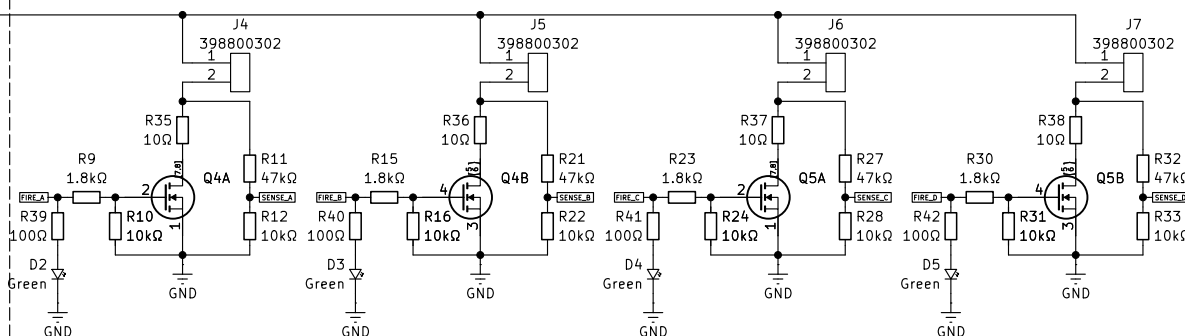
5V DC/DC regulator



$$V_{OUT} = ((R_{TOP} / R_{BOTTOM}) + 1) \times 0.8V_D = (V_{IN} - V_F) / R = ((2MΩ / 383kΩ) + 1) \times 0.8V = 5V - 2.2V / 200Ω = 14mA$$

$$P_{IN\text{ MAX}} = P_{5V} / \eta_{5V @ 3A} = ((5V \times 3A) / 0.9) = 16.67W$$

Pyrotechnic channel A,B,C,D



$$R_{LIMIT} = 10Ω$$

$$t = 100ms$$

$$I_{SENSE\text{ MAX}} = V_{MAX} / R_{TOT} = 8.4V / (47kΩ + 10kΩ) = 0.15mA$$

$$I_{MAX\text{ CH}} = V_{MAX} / R_{LIMIT} = 8.4V / (1.2Ω + 10Ω) = 0.84mA$$

$$I_{MIN\text{ CH}} = V_{MIN} / R_{LIMIT} = 6V / (1.8Ω + 10Ω) = 0.56mA$$

$$P_{LIMIT\text{ MAX}} = R_{LIMIT} \times I_{MAX}^2 = 10Ω \times 0.74A^2 = 5.48W$$

$$P_{PYRO\text{ CH MAX}} = V_{MAX} \times I_{MAX} = 8.4V \times 0.74A = 6.22W$$

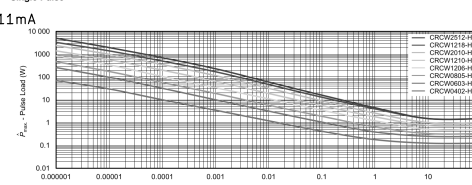
$$I_{LED} = (V_{IN} - V_F) / R = 3.3V - 2.2V / 100Ω = 11mA$$

$$I_{SENSE\text{ MIN}} = V_{MIN} / R_{TOT} = 6V / (47kΩ + 10kΩ) = 0.10mA$$

$$V_{SENSE\text{ MAX}} = V_{MAX} \times (R_{BOTTOM} / (R_{TOP} + R_{BOTTOM})) = 8.4V \times (10kΩ / (47kΩ + 10kΩ)) = 1.47V$$

$$V_{SENSE\text{ MIN}} = V_{MIN} \times (R_{BOTTOM} / (R_{TOP} + R_{BOTTOM})) = 6V \times (10kΩ / (47kΩ + 10kΩ)) = 1.05V$$

Single Pulse



Maximum pulse load, single pulse; applicable if $P \rightarrow 0$ and $n < 1000$ and $Q \leq Q_{max}$; for permissible resistance change equivalent to 8000 h operation



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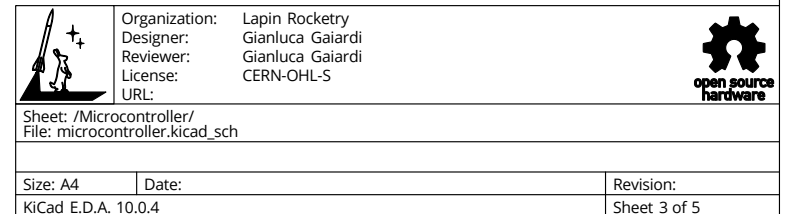
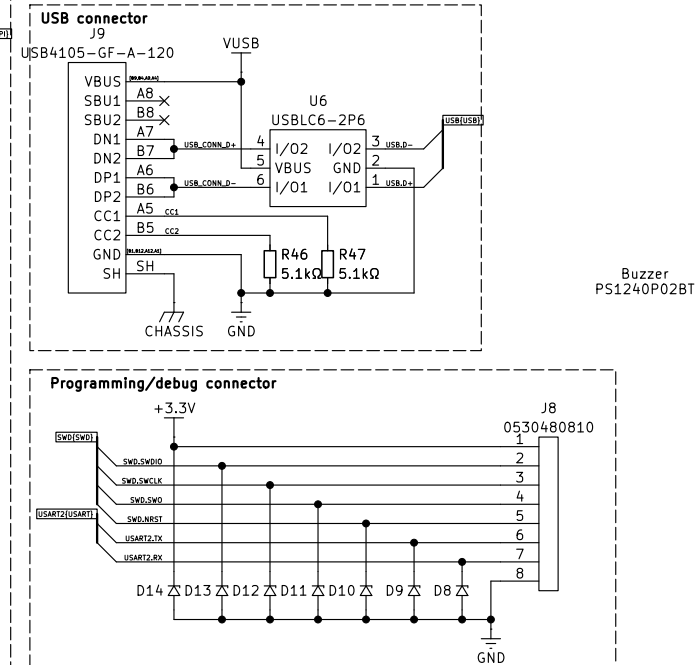
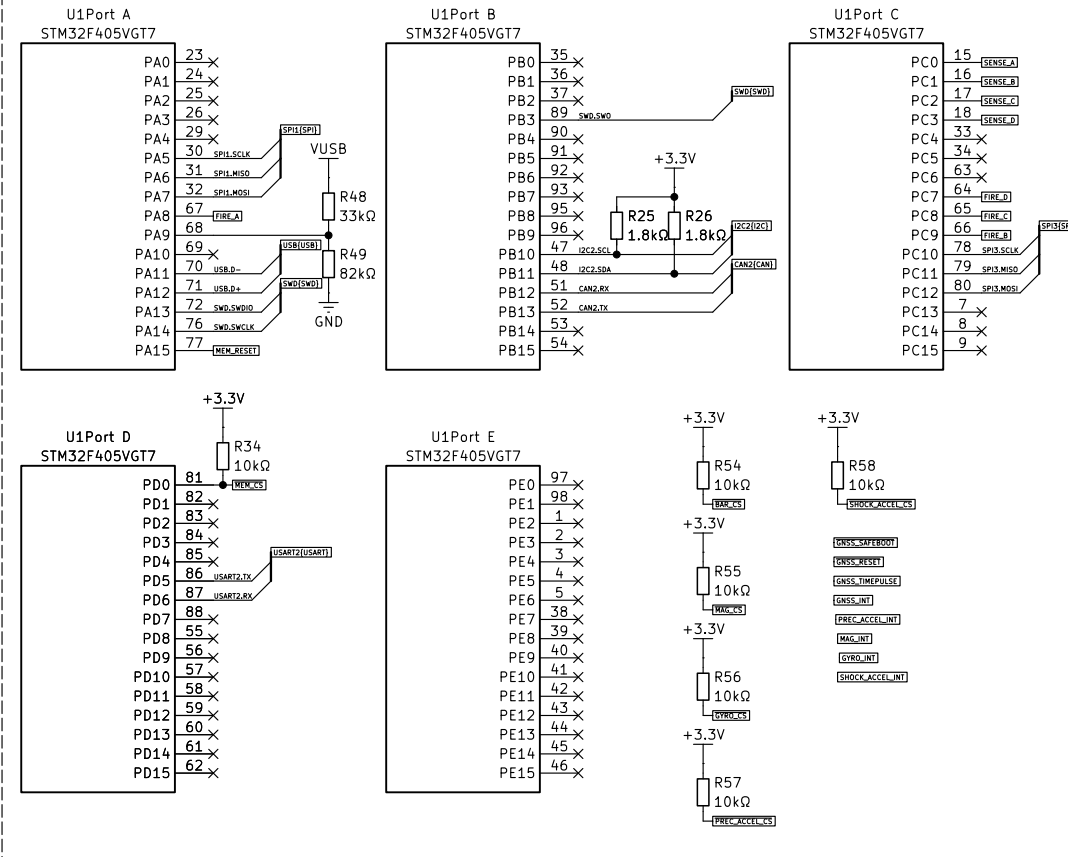
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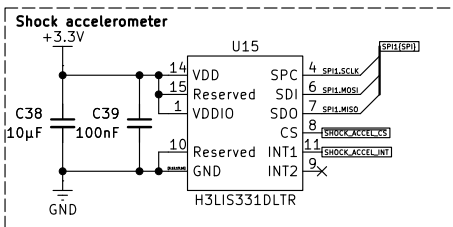
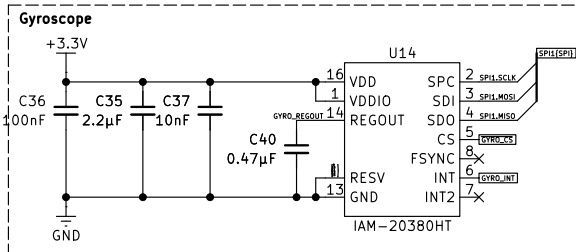
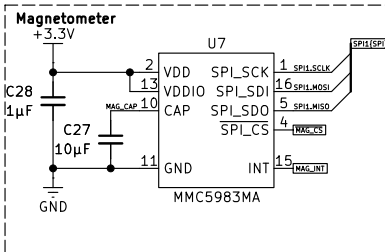
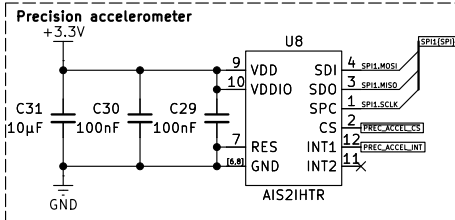
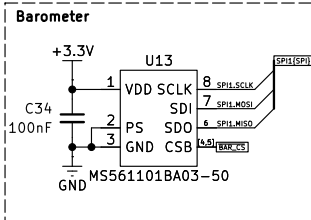
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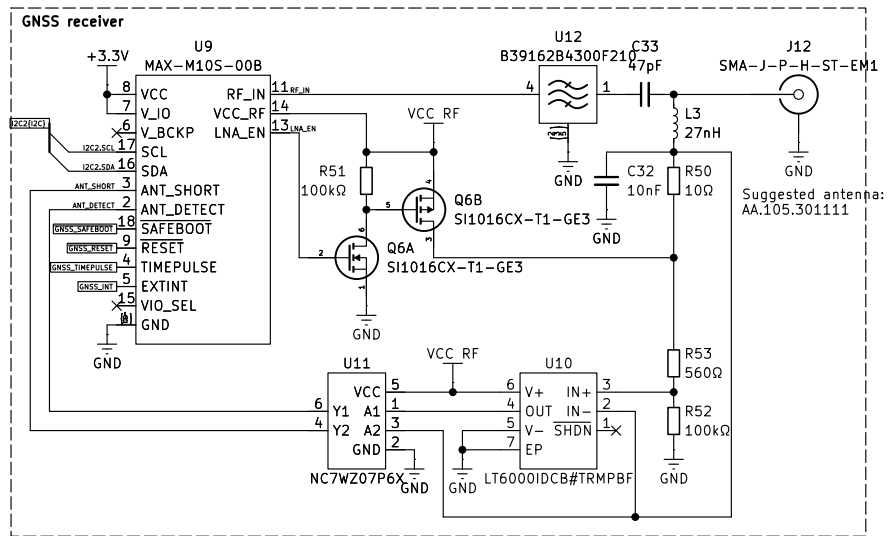


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Radio
CC1200
CC1201



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